**ClimOO-SDC2 design approach - 2026/07/20**

The SDC Ontology provides a relevant model of climate and climate change, primarily with a focus on Sustainable Development Goals (Davarpanah et al., 2023). Despite its relevance, the SDC Onto appears under-maintained and presents several shortcomings, including inadequate class axiomatizations, certain misleading labels, and an inconsistent hierarchy root. Here, we present our approach to reuse the SDC Onto model, update it, and overcome related shortcomings.

**Approach**

We forked SDC Onto to develop a novel version, named ClimOO-SDC2. ClimOO-SDC2 is the module of the ClimOO Ontology dedicated to climate change. ClimOO-SDC2 focuses on environmental features and related interconnections modeled in SDC Onto. We have not implemented SDC Onto models of, for instance, economic and social features or SDGs yet, as it is beyond the current scope of work of ClimOO. ClimOO-SDC2 extends SCO, aligning the SDC Onto model with primary existing BFO-compliant sustainability ontologies. We intend ClimOO-SDC2 as an initial starting point to establish a unique BFO-compliant representation of climate and climate change with respect to sustainability.

To construct ClimOO-SDC2, we proceed as follows (focus on OWL classes; we applied the same approach concerning the other OWL building blocks).

• If the class was not relevant, we did not import it.

• If the class was relevant and adequate, we directly imported it.

• If the class was relevant but inadequately axiomatized, we created a corresponding class that supports appropriate axioms but maintains the local identifier of the original SDC Onto class. We opted for this solution to offer a direct strategy of reference to the original version from which SDC2 has been forked. In the future, alternative, possibly more robust, strategies can be implemented. Exceptions: Given their centrality in the model, the following classes have been supplied with ClimOO local identifiers and annotated towards original SDC Onto classes:

- climate
- climate change
- (atmospheric) weather
- climate system process
- climate system component

• If the class was misleading, we did not import it, and we created adequate classes as needed. We supplied new classes with a ClimOO local identifier.

When revising classes, we complied with BFO and OBO-Foundry best practices. All SDC2 building blocks support the following namespace to ensure appropriate identification and fairness: https://www.w3id.org/climoo/climoo-sdc2#

• We introduced classes and axioms to expand the model with respect to underrepresented features, such as soil or molecular entities, as needed. See the OWL file directly.

*Addressing specific shortcomings*

- CCO has been updated several times; SDC Onto has not.

Strategy: we created ClimOO-SDC2 by importing from CCO 1.5, as needed. CCO has been updated to version 2.1. SC still imports from CCO 1.5. Aiming to conform with SCO, ClimOO-SDC2 currently reuses CCO imports in compliance with SCO. As soon as updates are made available for those resources, ClimOO-SDC2 will align accordingly.

- Absence of natural language definitions.

Strategy: We currently provide natural language definitions for core classes. We also included annotations to relevant external resources when appropriate. We foresee future implementations to provide natural language definitions for all OWL building blocks.

- Inconsistent hierarchy root.

Strategy: We revised SDC Onto hierarchies in compliance with BFO and SCO while constructing ClimOO-SDC2.

- Reusage of PATO representations that violate the BFO model, such as those concerning process properties, or that are inaccurate, such as those concerning increased and decreased dependent continuants.

Strategy: we revised SDC Onto models, either revising PATO-compliant building blocks or creating corresponding building blocks. The following classes have not been included:

- Decreased demand
- Decreased electricity demand for heating in winter
- Decreased heating oil demand for heating in winter
- Decreased natural gas demand for heating in winter
- Decreased wood demand for heating in winter
- Decreased area of mountain glaciers
- Decreased extent

- Decreased inland wetland extent
- Decreased soil moisture
- Decreased water availability
- Decreased freshwater availability
- Low water availability
- Decreased seasonal snow cover duration
- Increased aridity
- Increased availability
- Increased water availability
- Increased groundwater availability
- Increased surface water availability
- Increased water availability over Amazon
- Increased water availability over subtropics
- Increased temperature due to global warming
- Expanded total land area
- Increased soil moisture
- Water balance
- Event position
- Water efficiency
- Reduced snowpack
- Decreased water storage
- Increased sub-seasonal precipitation variability
- Increased seasonality of precipitation
- Increased frequency of concurrent extremes
- Increased frequency of concurrent extremes in crop producing areas
- Increased intensity of weather and climate extremes
- Event frequency
- Increased greenhouse gas concentration
- Increased plant water-use efficiency
- Increased flooding event frequency
- Increased frequency of coastal flooding
- Increased frequency of concurrent heatwaves and droughts
- Increased frequency of severe agricultural drought
- Increased frequency of severe ecological drought
- Increased frequency of severe meteorological drought

    - Increased heavy precipitation frequency
    - Increased seasonality of precipitation over Amazon
    - Increased seasonality of precipitation over subtropics
- Label ambiguity of certain classes (all class labels have been renamed in lowercase).

Strategy: We added comments and explanations to reduce misleadingness. We fixed labels as needed.

Addressed classes:

- Cryosphere material entity classes
- Body of water class.
- Ocean part class.
- Material substance classes.
- Storage classes.
- Water classes.
- Reserve class.
- Sink classes.
- Variability classes.
- Runoff intensity classes.
- Seasonality classes.
- Water-use efficiency classes.
- The water cycle variability class.
- The thermal expansion class.
- The exchange of heat with the atmosphere class.
- The oscillation class.
- The exchange of heat with the atmosphere class.
- The ocean current class.
- The groundwater recharge class.
- The stomal regulation class.
- The melting class.
- The freezing class.
- Evaporation classes.
- The condensation class.
- Loss classes.
- The surface water runoff.
- The infiltration class.
- The loss of ice mass class.

    - The glacial mass loss class.
    - The evapotranspiration class.
    - The shift in annual precipitation class.
    - The shift in seasonal precipitation class.
    - The change in water reservoir class.
    - Leaf stomata closure class.
    - The warming class.
- Inadequate class axiomatizations and misleading representations.

Strategy: see the general approach above.

Addressed classes (see the OWL file for more details):

- Cryosphere material entity classes: deleting the role axiomatization. Adding axiomatizations to some subclasses.
- Ocean part class.
- River and lake classes.
- Flood classes: we created adequate process classes.
- Lake and river classes.
- Material substance classes.
- Water supply classes: classified as material entity, while (at least intuitively) referring to processes. No definition is provided, limiting interpretation. We created a water supplying process class and a general portion of supplied water class.
- Storage classes. Reclassified as not all storage materials are hydrological.
- The resilience class.
- The aridity class.
- Deficit classes.
- Anthropogenic activity class and subclasses: Several of the subclasses are not including only instances of human-driven processes. We reclassified them accordingly.
- Carbon dioxide removal on land and solar radiation modification classes.
- Act classes.
- Flow and water flow classes.
- Runoff classes.
- The leaf stomata closure class.
- Atmospheric physical process classes.
- Event classes.

- Climate system process classes.
- Change classes.
- The dangerous heat threshold class.

Bibliography

Davarpanah, A., Babaie, H., & Dhakal, N. (2023). Semantic modeling of climate change impacts on the implementation of the U.N. sustainable development goals related to poverty, hunger, water, and energy. *Earth Sci. Inf.*, *16*(1), 929–943.